Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 09: Statistics I

Digital for Executive

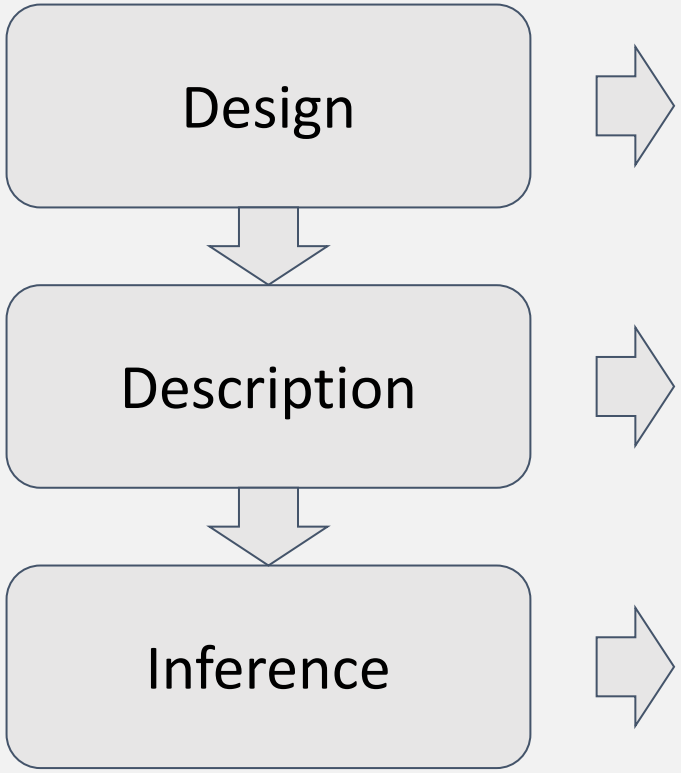
Outline

- Introduction to Statistics
  - What is Statistics
  - Data
  - Step by step statistics
- Design Study
- Descriptive Statistics

What is Statistic?

- Statistic could be described as methodology for collecting, analyzing, interpreting, and drawing conclusion from data.
- Its ultimate goal is translating data into knowledge which can help us making any decision.
- It is a very broad subject, with many application on various field:
  - Experimental Design, A/B Testing for new web apps design
  - Survey, Predicting Election Using an Exit Poll / Quick Count
  - Research, Making Conclusion in Medical Research Studies
  - Quality Control for any product in a factory, etc

Step by Step Statistics



Design: Planning and carrying out research studies also data collection

Description: Summarizing and exploring data

Inference: Making predictions and generalizing about phenomena represented by the data

Step by Step Statistics

Design

- Formulate Research Problem
- Define Population and Sample
- Data Collection

Description

Descriptive Statistics
- Graphical Summary
- Numerical Summary
- Table Summary

Inference

- Use appropriate Method to solve the problem

Type of Statistics

Design Step

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Logos
Extraits d'avis
Toutes les pages connues
Introuvable (404)
État : Exclues
EN SAVOIR PLUS
Pages concernées
52
Exemples
URL
MacBook Air

Design Statistics

- Data
- Type of Study
- Population and Sample
- Randomness
- Sampling
- Experimental

Data

Data is individual units of information

	Name	Author	User Rating	Reviews	Price	Year	Genre
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350	8	2016	Non Fiction
1	11/22/63: A Novel	Stephen King	4.6	2052	22	2011	Fiction
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979	15	2018	Non Fiction
3	1984 (Signet Classics)	George Orwell	4.7	21424	6	2017	Fiction
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665	12	2019	Non Fiction

Observation/Unit (Books)

Data source: https://www.kaggle.com/sootersaalu/amazon-top-50-bestselling-books-2009-2019

Variables

Variables

A characteristic that varies from one person or thing to another is called a variable. Ex: Height, Weight, Eye Color, etc.

Type of variable

Variable

- **Quantitative**
  - Take On Numerical Value. Sensible to add and subtract
  - **Continuous** — Percentage, height, weight, age. etc
  - **Discrete** — Number of family member, Number of Child, Number of pets in a household, etc
- **Qualitative**
  - Observation belong to one set of distinct category
  - Gender, Satisfaction Level, Education Level, etc

Why do we care about type of variable ?

In Practice Data Analysis Depends on Type of Variable, example:

- should we use classification or regression when predicting whether a customer will leave or not
- should we use average to describe Number of pets in a household, etc

Scale of Measurement

Nominal: Qualitative variables that have two or more categories, but did not have intrinsic order. Ex: Type of fruit (Apple, Banana, Grape), Type of properties(Commercial or Residential), Own a house (Yes or No). Special type of **Nominal** scale when only has two category is **Dichotomous or Binary**.

Ordinal: Qualitative variables that have two or more categories, but the categories could be ranked or ordered. Ex: Satisfaction level (Satisfied, Normal, Not Satisfied), Education Level (SD, SMP, SMA).

Interval: Quantitative variable which their central characteristic could be measured along continuum value and have numerical value. Multiplication or division are not sensible. Ex: Temperature, 40C is not as hot as 2 x 20C.

Ratio: Interval variables, but with the added condition that 0 (zero) of the measurement indicates that there is none of that variable. Multiplication or division are sensible. Values from ratio variable usually is greater than 0 or lower than 0. So, temperature measured in degrees Celsius or Fahrenheit is not a ratio variable because 0C does not mean there is no temperature. Ex: Weight 0 kg meaning the unit doesn't exist, 2 x 4 kg is equal to 8 kg. However, temperature measured in Kelvin is a ratio variable as 0 Kelvin (often called absolute zero) indicates that there is no temperature whatsoever.

Scale of Measurement Summary

Scale	Compare	Distance	Order	Classify	Zero	Multiplication or division
Nominal	-	-	-	v	-	-
Ordinal	-	-	v	v	-	-
Interval	-	v	v	v	Non-absolute	Not Sensible
Ratio	v	v	v	v	Absolute	Sensible

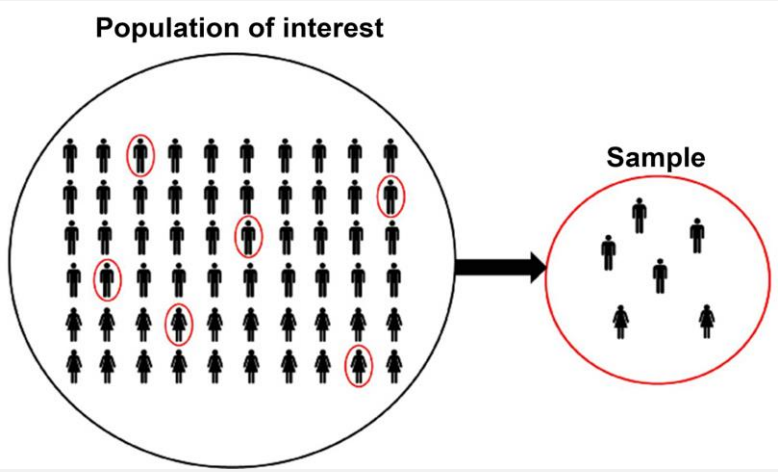
Experimental Study and Observational Study

Experimental Study. A researcher conducts an experimental study, or more simply, an experiment, by assigning subjects to certain experimental conditions and then observing outcomes on the response variable (or variables). The experimental conditions, which correspond to assigned values of the explanatory variable, are called treatments.
Ex, **A/B Testing** of new web design to increase the conversion rate.

Observational Study. In an observational study, the researcher observes values of the response variable and explanatory variables for the sampled subjects, without anything being done to the subjects (such as imposing a treatment).
Ex, **sample survey** for quick count.

Population and Sample

- Population can be characterized as the set of individual persons or objects in which an investigator is primarily interested during the research.
- Set of individual or object observed as representation of the population is called sample
- Sample should represent the population.
- We learn about the population only from the samples because of so many restriction like time, cost, source etc.



Sampling

- **Sample** should **represent** the **population** of interest
- A method to choose appropriate sample is called **sampling**
  - sample obtained randomly
- Be cautious of Sampling bias

Randomness

Let's analyze data gathered from a hospital.

	Heart Disease	No Heart Disease	Total
Age > 50	57.8 %	42.2 %	100%
Age < 50	88.8 %	11.2 %	100%

- Notice that young people have higher proportion suffer from heart disease 88.8%.
- Is that make sense that younger people have tend to have higher risk of suffering from heart disease ?

Data from table are **not collected randomly**. The reason why younger people have higher proportion of heart disease may be because their awareness to check their health is lower than older people. **Younger people** only **check** only **if** they start to **feel** something **wrong** with their body while **older people** because of their **awareness** regardless there is something wrong with their body or not. **Data** is still part of population but **not representative**.

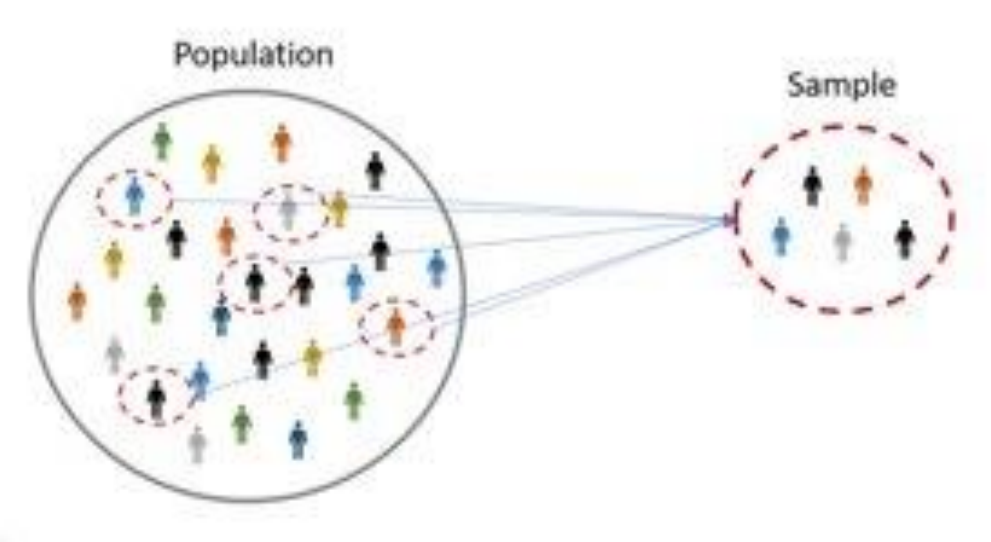
Why do we need sample ?

- When a Doctor want to test your blood, Should he/she take all of your blood ?
- When your Mom is cooking, should she eat all the cook or eat only small part of it ?

Things to Consider

- Resource
- Time
- Cost

Sample Size

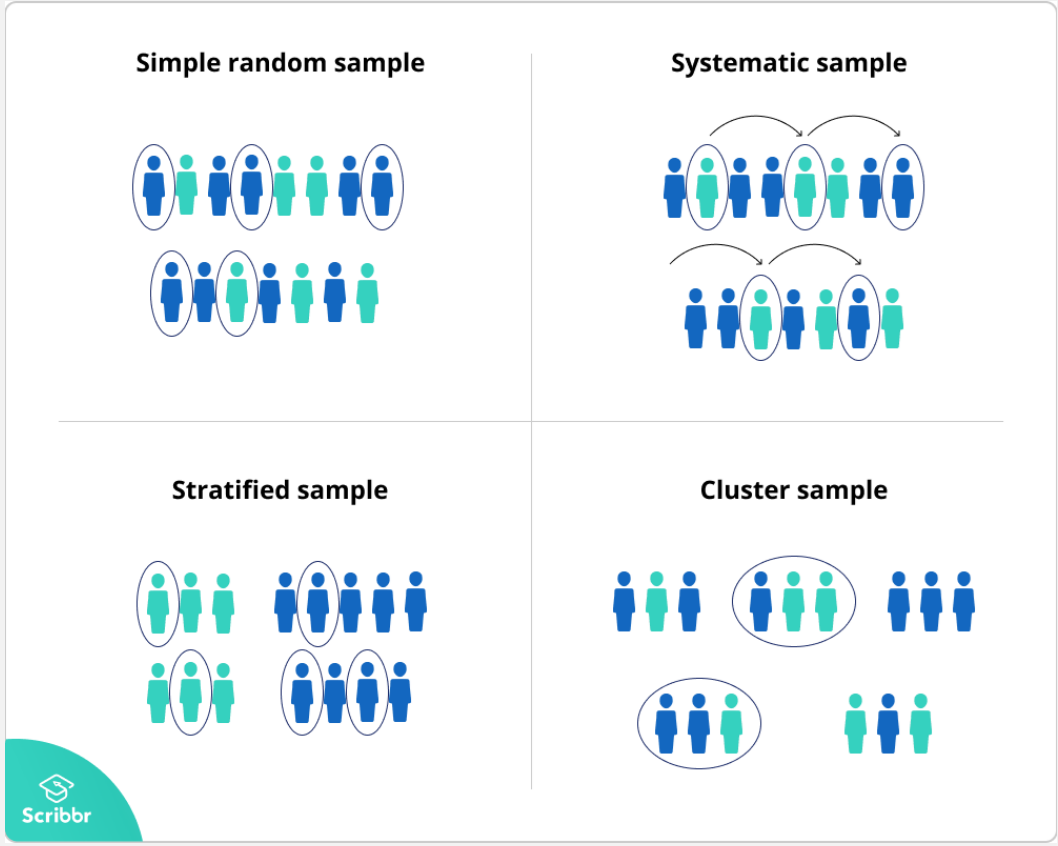


Let N be the population size and the margin of error e denotes the allowed probability of committing an error in selecting a small representative of the population.

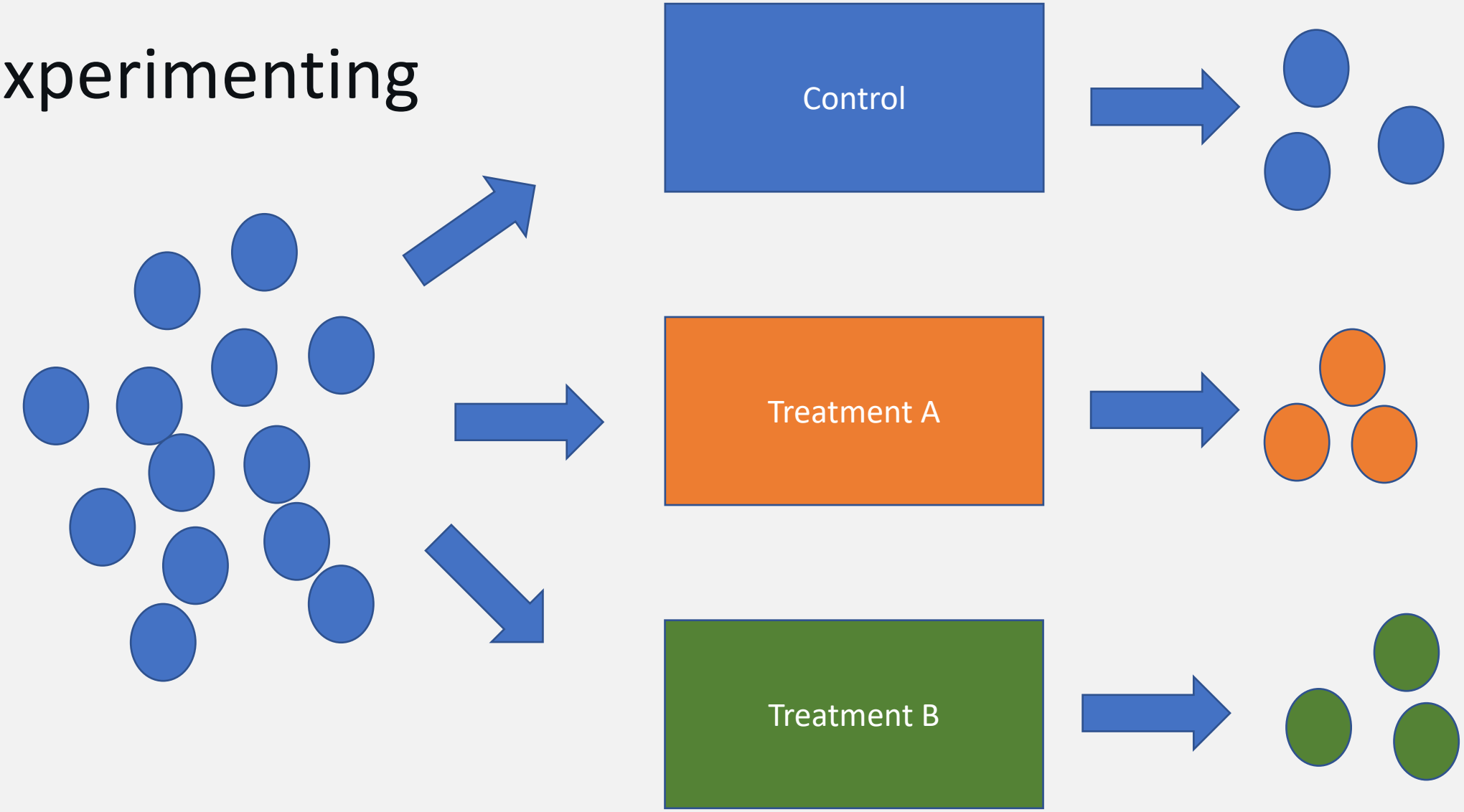
The sample size n can be obtained by the formula

$$n = \frac{N}{1 + Ne^2}$$

How to take sample?



Experimenting



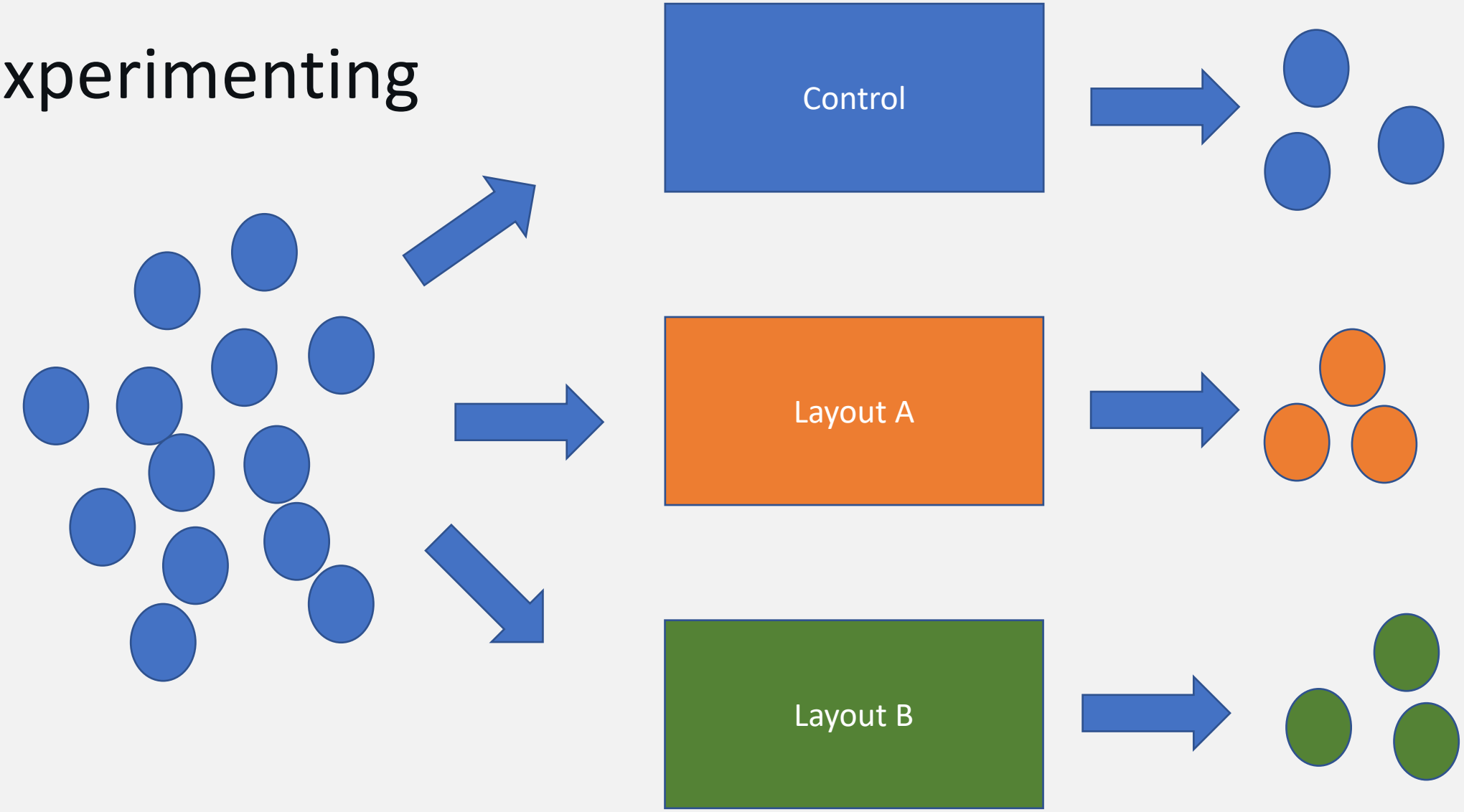
Example of an Experimentation

Marketplace want to test whether their new app design help the company to increase the conversion rate.

1. Six months ago, the company randomly selected 240 newly signed up users and assigned 122 of them to the control and each 118 to the new designs.
2. control group went to current design (layout A)
3. The 240 users represent just a small portion of the app's total users (sample).
4. Subject on this experiment is people.
5. By Assigning randomly, we minimize effect of covariate. (ex gender, age, job, etc) for each group.
6. In the end we want to compare the conversion rate.

	Layout A	Layout B
Visitors	122	118
Customers	22	25
Conversion %	18.0%	21.2%

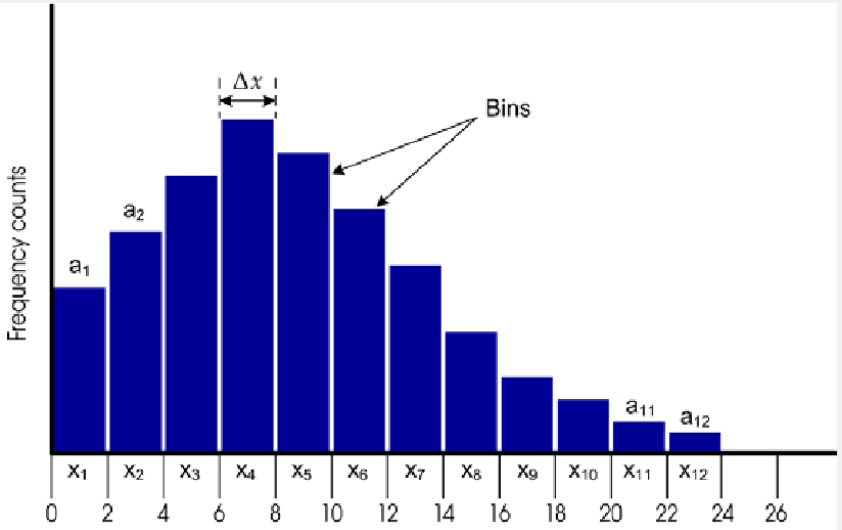
Experimenting



Descriptive Statistics

Descriptive Statistic

- To summarize and explore the data
- Descriptive statistic includes the construction of graphs, charts, tables, and calculation of various descriptive measures such as averages, variation, and percentile.




```
tips = sns.load_dataset('tips')
```

```
tips.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

Numerical Summary

There are generally 2 types of numerical summary:

- **Measures of Central Tendency.** It is the way of describing the central position of a frequency distribution for a group of data. We can describe it by using, for example Mean, Median, Mode

- **Measures of Spread.** It is the way to summarize the group of data by describing how spread the data are. We can describe it by using, for example Range, Quartile, Variance, and Standard Deviation

Measure of Central Tendency

There are three most common way to describe the central measurement of the frequency distribution:

- **Mode**: Value of a qualitative or a countable quantitative variable where the frequency is occurring the most.
- **Median**: The middle value in the ordered list. If the number of observation is odd, then the sample median is the observed value exactly in the middle. If the number of observation is even, then the sample median is the number halfway between the two middle observed values in the ordered list. In either case, the sample median position is at n+1/2 when n is the number of observation
- **Mean**: The sum of observed values in a data divided by the number of observations. The most commonly used measure of center for quantitative variable.

Which measurement to choose?

- **Mode** should be used when calculating the measure of center for the qualitative variable
- **Mean** is the proper measure of center if dealing with the quantitative variable with symmetric distribution (often bell shaped)
- **Median** is the good choice if the quantitative variable have a skewed distribution. We do not used mean in this case, because mean could be highly influenced by an observation that falls far from the rest of the data (outlier)

Measure of Central Tendency Example

Patient	gender	age
Andrew	Male	22
Jacob	Male	22
Ros	Female	23
Andersen	Male	23
Lina	Female	29
Robert	Male	24
Jack	Male	27
Annie	Female	28

- Mode
  - Gender : Male
  - Age : 22 and 23
- Mean
  - Age

    (22 + 22 + … + 28) / 8 = 24.75
- Median

  22, 22, 23, **23, 24**, 27, 28, 29

  23.5

Measure of Spread

- Another important aspect of descriptive study is numerically measuring the extent of variation around the center. Two dataset of the same variable may possess similar position of center but remarkably different with respect to variability.
- Most frequently used measures of variation; the sample range, the sample interquartile range, and the sample standard deviation

Range

- The sample range is obtained by computing the difference between the largest observed value of the variable and the smallest one

Range = Max – Min

- Range is overly sensitive to extreme value

Variance and Standard Deviation

- The sample standard deviation is the most frequently used measure of variability. It can be considered as a kind of average of the square deviations of observed values from the mean of the variable.

$$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

variance

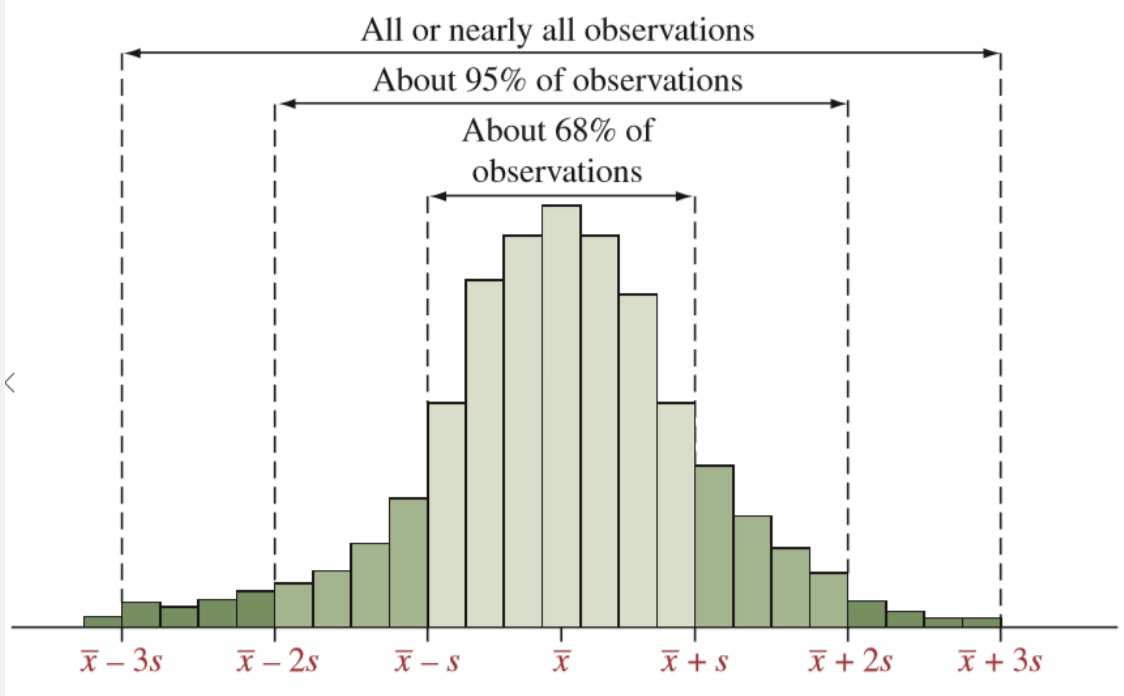
***s* = Sample Standard Deviation**
***X_i* = Each value of dataset**
***$\bar{x}$* = Mean of the dataset**
***N* = Sample size**

Standard Deviation Properties

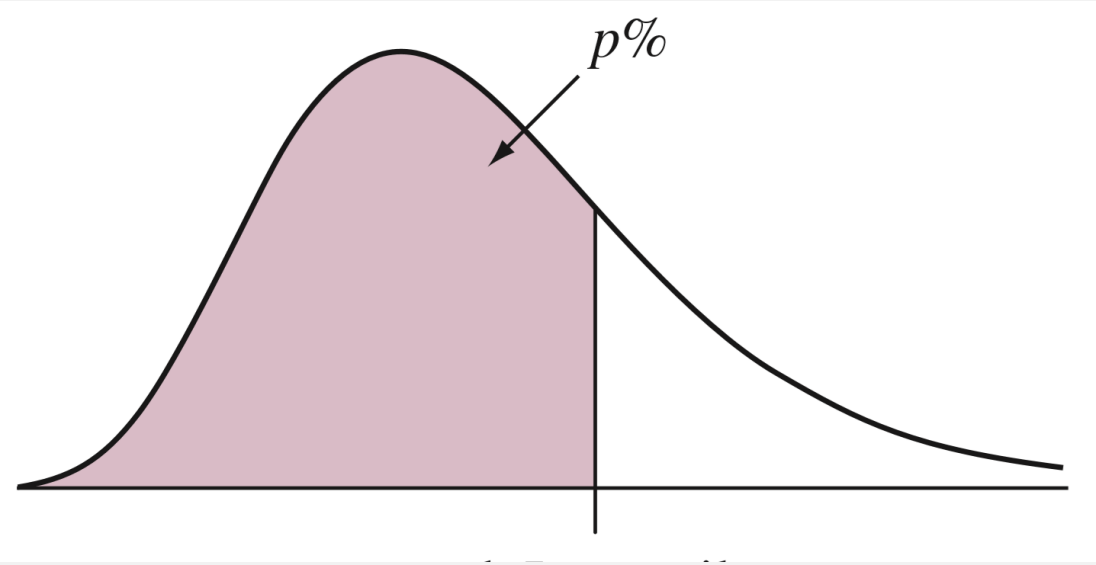
- The more variation in the observed value, the larger the standard deviation for the variable observed.
- Standard Deviation is greatly affected by a few extreme observation (usually outlier).

When a distribution of Data is bell shaped, then approximately:

- 68% of observation fall between $\bar{x} - s$ and $\bar{x} + s$
- 95% of observation fall between $\bar{x} - 2s$ and $\bar{x} + 2s$
- 99.7% of observation fall between $\bar{x} - 3s$ and $\bar{x} + 3s$
- $s \approx$ Range / 4 = (Max – Min) / 4

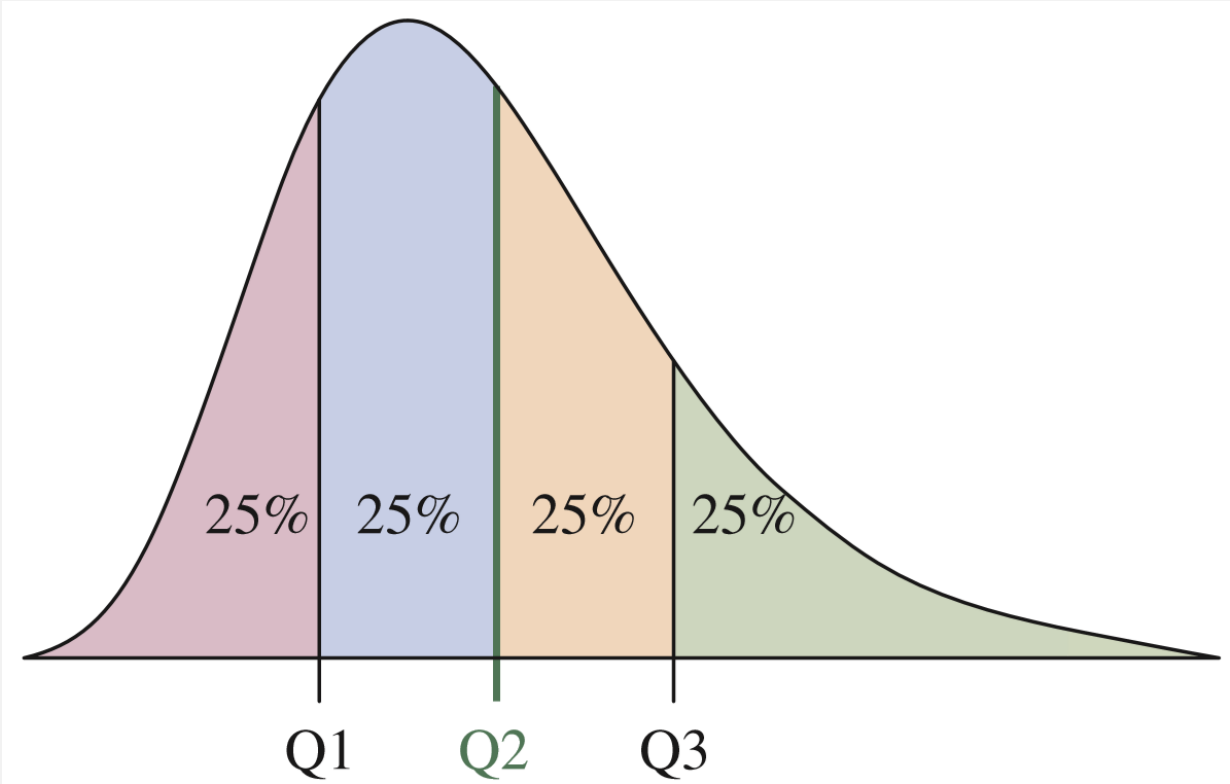


Percentile



The ***p*th Percentile** is a value such that *p* percent of the observation fall below or at that value.

Quartile



Quartile is a special case of **percentile**.

- Q1 is percentile 25
- Q2 is percentile 50 or also known as median
- Q3 is percentile 75

IQR and Outlier

Interquartile range (IQR) is the distance between Q1 and Q2:

IQR = Q3 - Q1

IQR can be used as replacement for standard deviation when data is normally distributed because standard deviation is very sensitive to extreme value

S = 1.34898 x IQR

IQR is used to detect the potential outlier

An observation considered outlier if the value

- below Q1 - 1.5 x IQR
- above Q3 + 1.5 x IQR

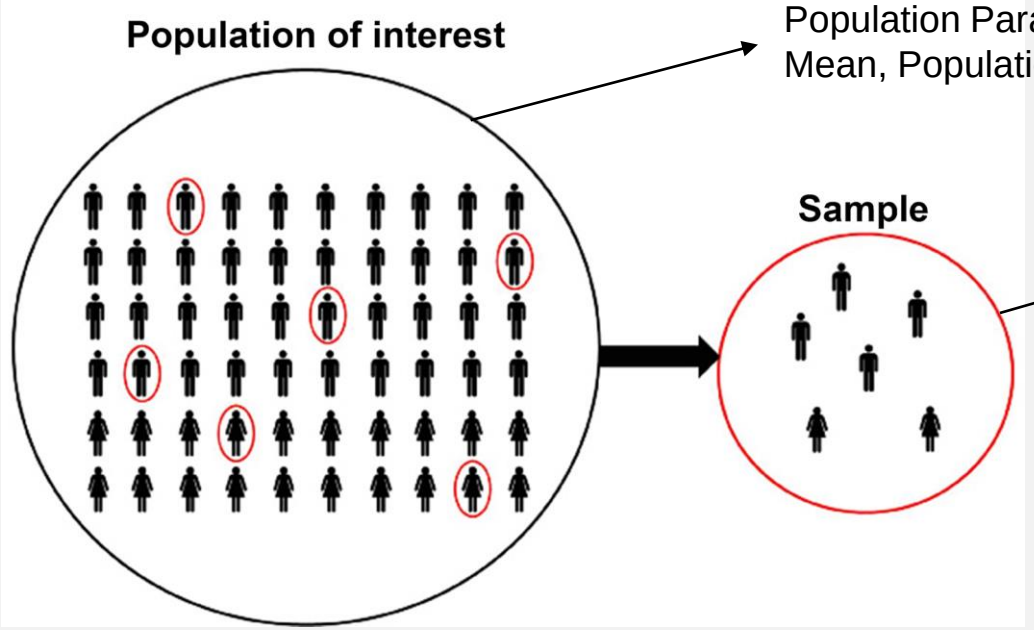
Measure of Spread Example

Patient	gender	age
Andrew	Male	22
Jacob	Male	22
Ros	Female	23
Andersen	Male	23
Lina	Female	29
Robert	Male	24
Jack	Male	27
Annie	Female	28

- Range
  - Age
    29 - 22 = 7
- Standard Deviation
  - Age
    s = 2.63391
- Q2
  22, 22, 23, **23, 24**, 27, 28, 29
  23.5
- Q1 22.75
- Q3 27.25
- IQR 4.5

Statistics and Parameter

- **A parameter** is a numerical summary of the population. **A statistic** is a numerical summary of a sample taken from the population.
- Population parameter are unknown and sample statistic used to make inference about it



Population Parameter such as Population Mean, Population Median, etc.

Statistic sample such as Sample Mean, Sample Median, etc.

Table Summary

- Frequency table
  - for numerical
  - for categorical
- cross tabulation

Frequency Table

Frequency Table for categorical variable

Day	Visitor Count
Saturday	87
Sunday	76
Thursday	62
Friday	19

Frequency Table for numerical variable

Tip Range ($)	Visitor Count
0 - 2.5	108
2.5 - 4	95
4 - 5.5	29
5.5 - 7	9

Cross Tabulation / Contingency Table

Cross Tabulation : Frequency

Day	Visitor Count (Male)	Visitor Count (Female)
Saturday	87	32
Sunday	76	9
Thursday	62	28
Friday	19	18

Cross Tabulation : Percentage

Day	Visitor Count (Male)	Visitor Count (Female)	Total
Saturday	73.1	26.9%	100%
Sunday	89.4%	10.6%	100%
Thursday	68.8%	31.1%	100%
Friday	51.3%	48.6%	100%

Graphical Summary

Numerical :

- Histogram
- Boxplot
- Scatterplot, etc

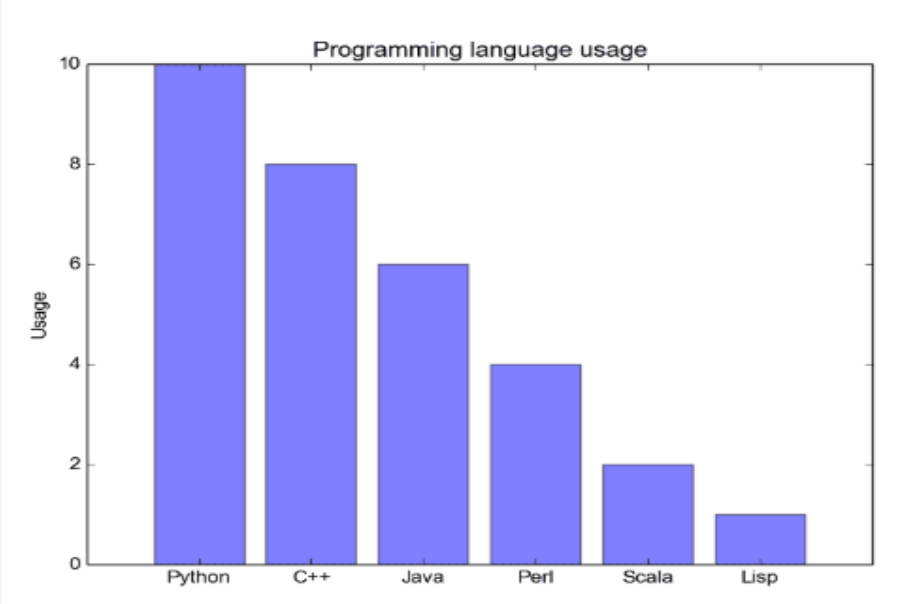
Categorical

- Pie chart
- Barchart, etc

Both numerical and Categorical:

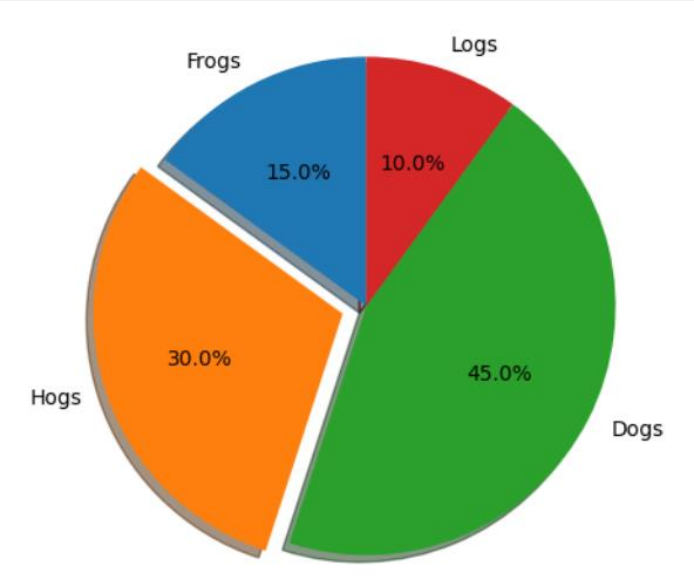
- Barplot
- Boxplot

Bar chart



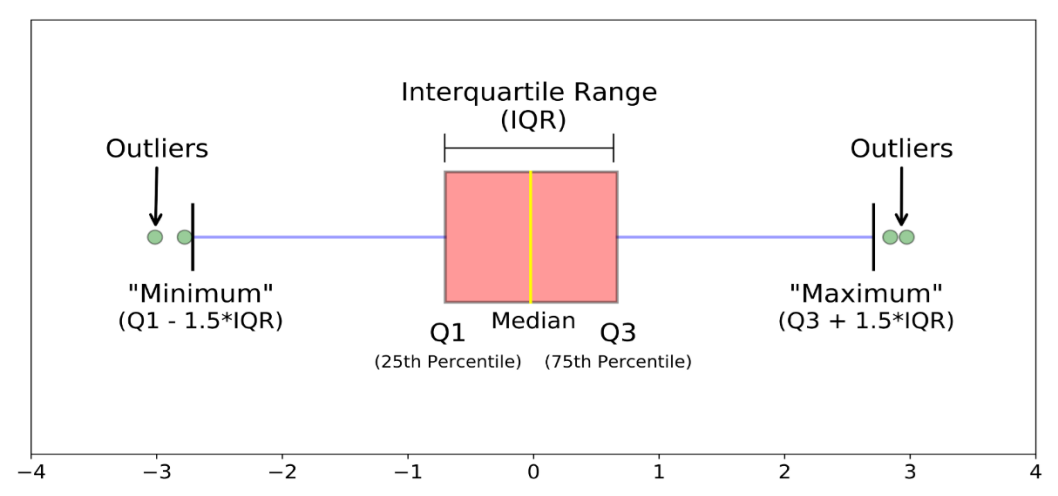
- Represents **categorical data** with rectangular bars. Each bar has a height corresponds to the value it represents.
- Each category can be overlap or non overlap.
- Can be used to see composition (nonoverlapping) or comparison (overlapping)

Pie chart



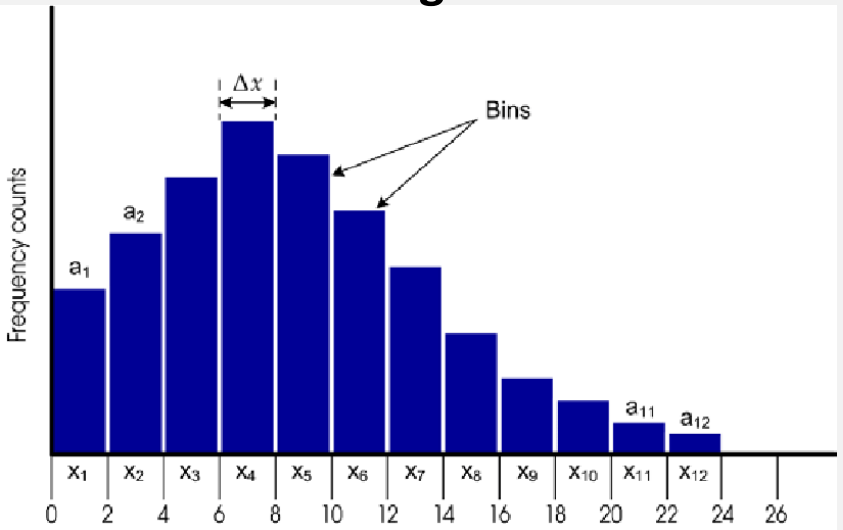
- A circular plot, divided into slices to show numerical proportion of the categorical data. They are widely used in the business world.
- Each category are non-overlapping
- Main purpose is composition
- Not recommended if there are too many categories

Boxplot



- Box plot, also called the box-and-whisker plot: a way to show the **distribution of values based on the five-number summary**: minimum, first quartile, median, third quartile, and maximum.
- Can be used to detect anomaly data/outliers

Histogram

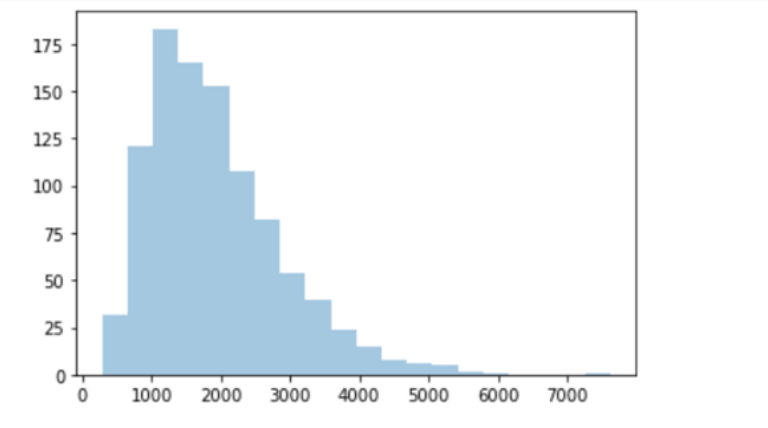


- **Histogram** is an accurate representation of the **distribution of numeric data**.
- A histogram is a graph that uses bars to portray the frequencies or the relative frequencies of the possible outcomes for a quantitative variable.

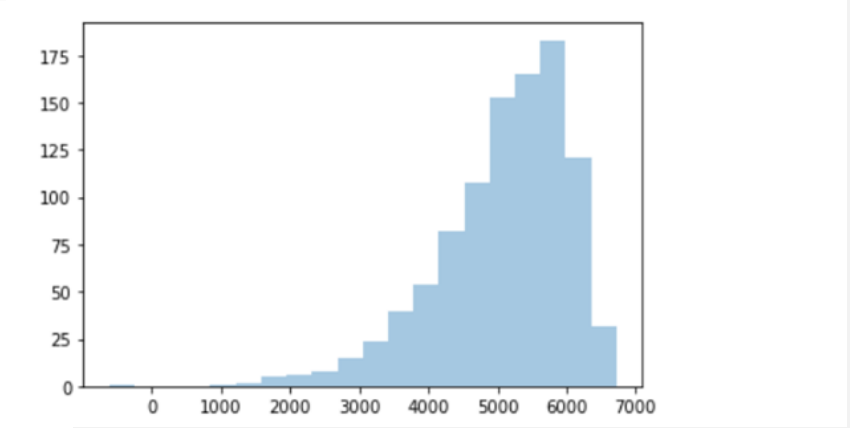
Histogram

Using histogram we can see how data spread.

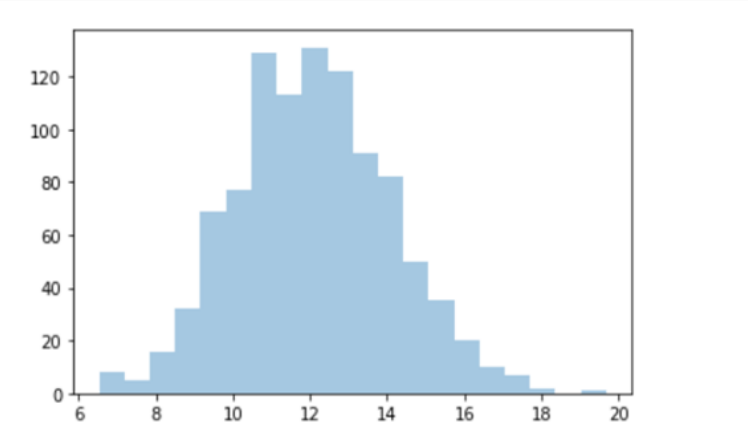
Right Skewed



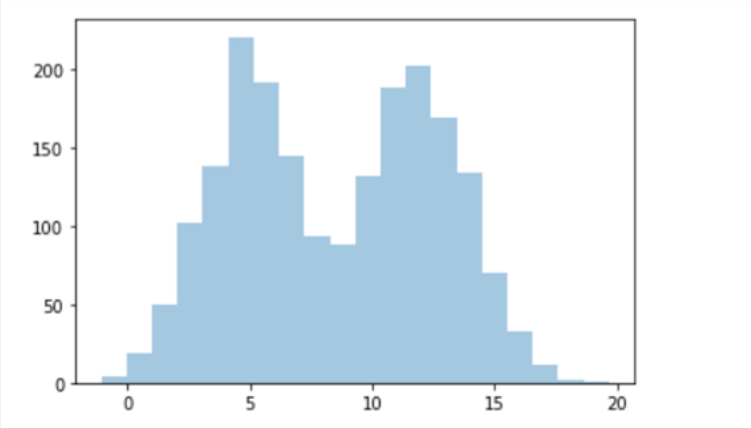
Left Skewed



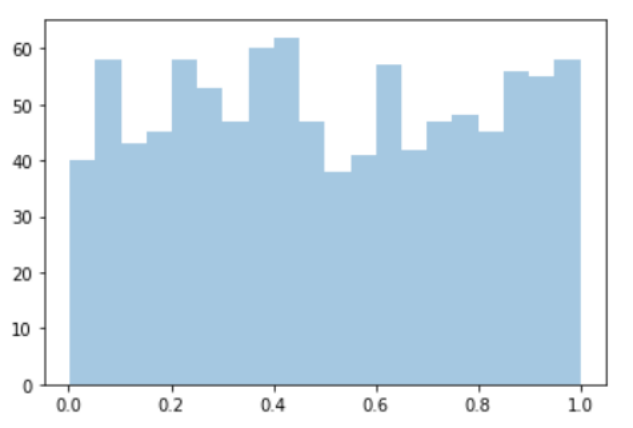
Bell-shaped



Bimodal



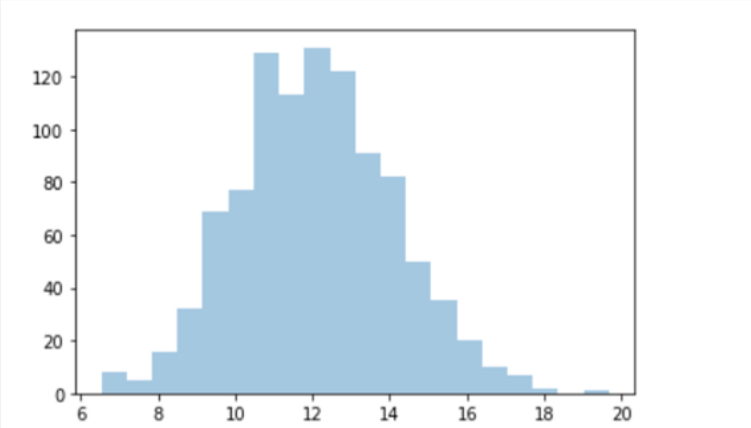
Uniform

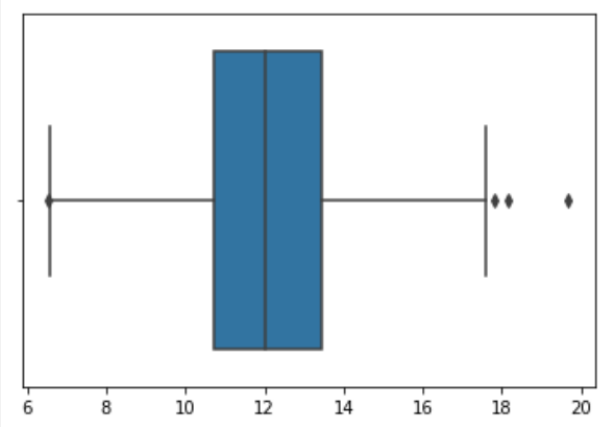


Boxplot

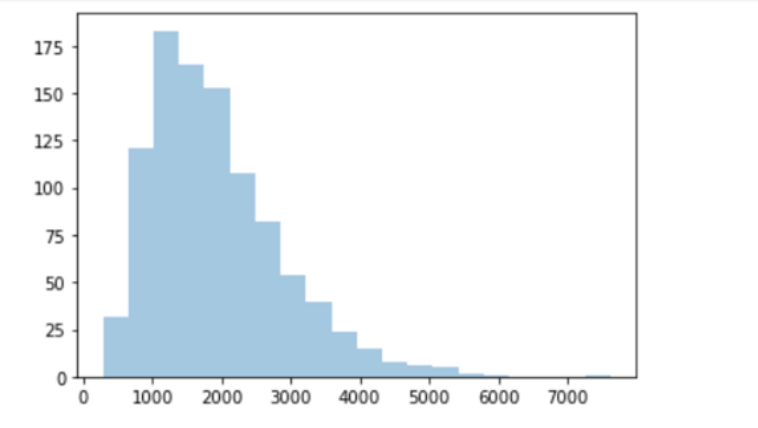
Using boxplot we can detect outliers

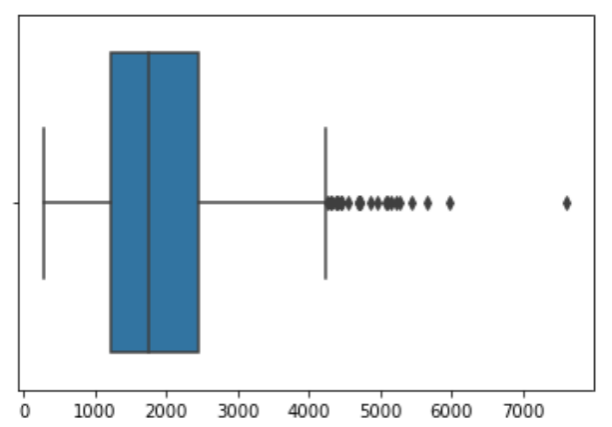
Symmetric or Normally Distributed



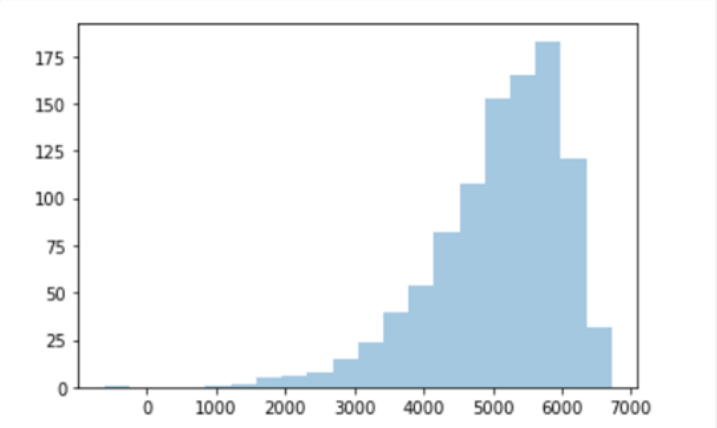


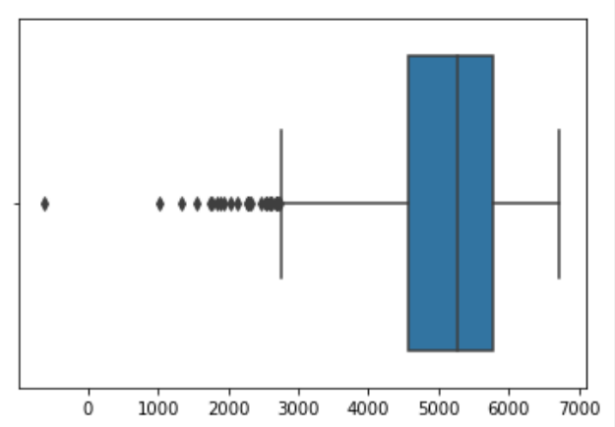
Right Skewed



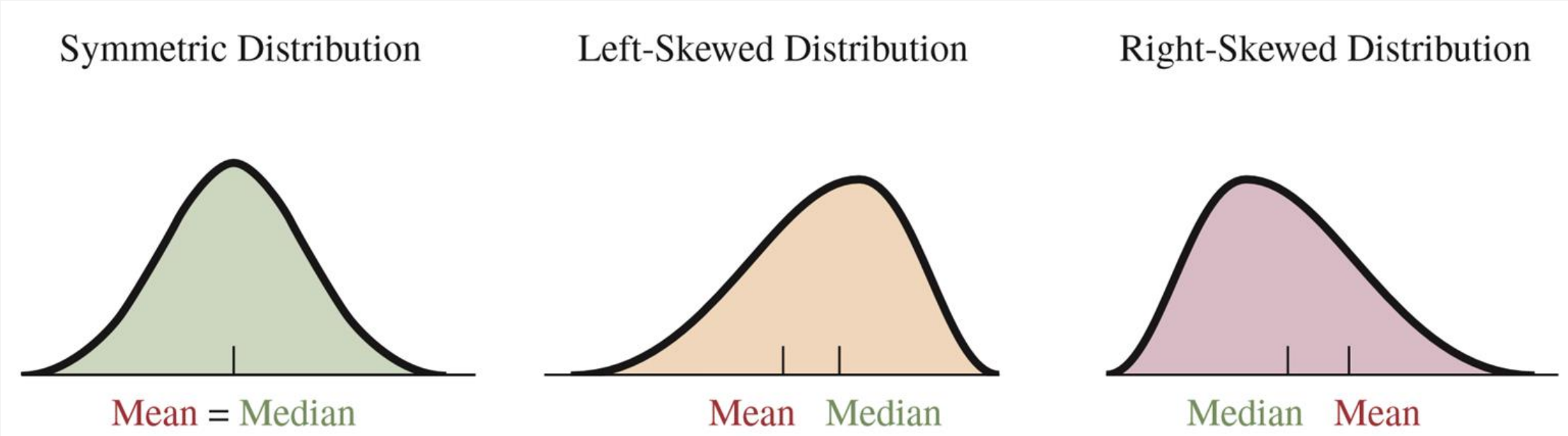


Left Skewed

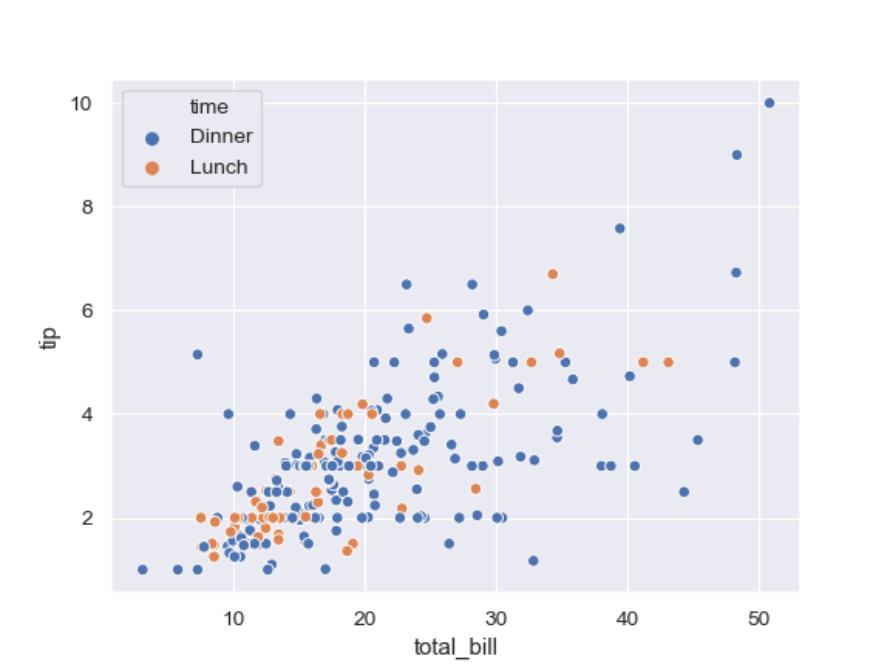




Median vs Mean

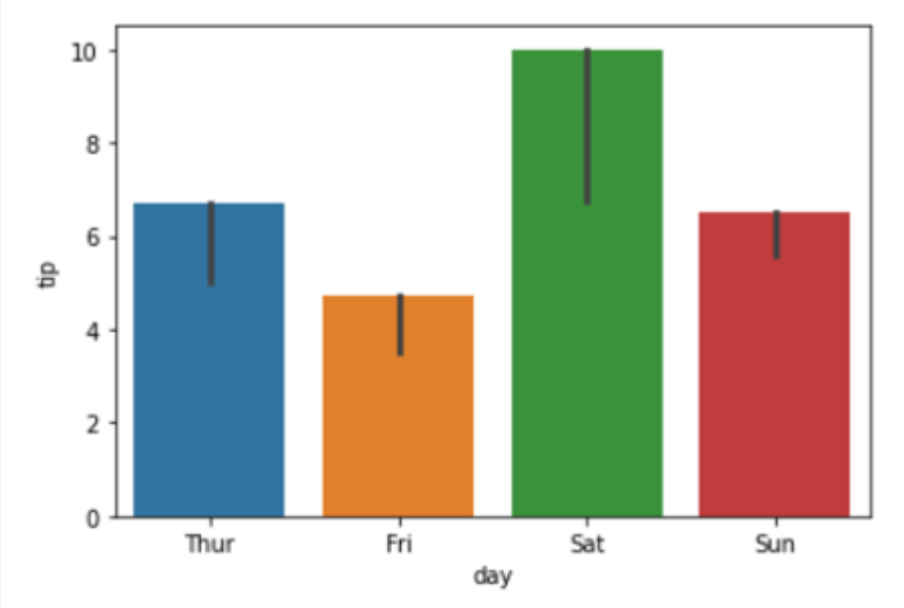


Scatterplot



- this type of plot shows **all individual data points**. Here, they aren't connected with lines.
- Each data point has the value of the x-axis value and the value from the y-axis values.
- This type of plot can be used to display **trends or correlations**.
- In data science, it shows relationship between **two numerical variables**.

Barplot



- **Barplot** is a general plot that allows you to aggregate some values in the categorical data based on some function (mean, sum, min, max, std, etc)
- In data science, it shows composition and relationship between **a numerical variables** and **a categorical variables**.

Thank You!

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